

Math 170E

Week 6

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1 Normal Random Variables

Example 1. (PSI 3.2.24) Let the random variable X be equal to the number of days that it takes a high-risk driver to have an accident. Assume that X has an exponential distribution. If $\mathbb{P}(X < 50) = 0.25$, compute $\mathbb{P}(X > 100|X > 50)$.

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{2}}{2}$ (D) $\frac{3}{4}$ (E) $1 - \frac{\sqrt{2}}{2}$

Example 2. (PSI 3.3.1-modified) If Z is $N(0, 1)$, find the following probabilities using standard normal table:

1. $\mathbb{P}(Z \leq 1.2)$

2. $\mathbb{P}(Z \leq 0.87)$

3. $\mathbb{P}(0.53 < Z \leq 2.06)$

4. $\mathbb{P}(Z \leq -1)$

5. $\mathbb{P}(-0.79 \leq Z < 1.52)$

6. $\mathbb{P}(Z > -1.77)$

7. $\mathbb{P}(Z > 2.89)$

8. $\mathbb{P}(|Z| < 1.96)$

9. $\mathbb{P}(|Z| < 2)$ (A) 0.453 (B) 0.671 (C) 0.881 (D) 0.954 (E) 0.98

(see class)

Example 3. (PSI 3.3.7) If X is $\mathcal{N}(3, 4)$, find $\mathbb{P}(|X| < 4)$

- (A) 0.13 (B) 0.25 (C) 0.36 (D) 0.56 (E) 0.69

Example 4. (PSI 3.3.7) If X is $\mathcal{N}(650, 400)$, find

1. $\mathbb{P}(600 \leq X < 660) \approx$ (A) 0.68 (B) 0.72 (C) 0.83 (D) 0.94 (E) None
 2. A constant $c > 0$ such that $\mathbb{P}(|X - 650| \leq c) = 0.9544$
 (A) 12.1 (B) 24.7 (C) 36 (D) 39.2 (E) 44.9

(F) 40

Example 5. (Old Quiz Problem) Let X be the standard normal $\mathcal{N}(0, 1)$. Find the pdf of $Y = |X|$.

$$P(Y \leq y) = P(|X| \leq y) = P(-y \leq X \leq y) = F_X(y) - F_X(-y)$$

$$f_Y(y) = \frac{d}{dy} (F_X(y) - F_X(-y)) = f_X(y) - (-f_X(y)) = 2f_X(y)$$

Example 6. (Old MT Problem) Given a constant c , suppose X is a continuous RV with pdf

$$f(x) = c(x + 3), \quad -2 < x < 1$$

$$= \frac{c}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}$$

for $y \geq 0$

1. Find c .
 2. Let $Y = |X + 1|$. Find the support and pdf of Y .

Example 7. (Old MT problem) Let us consider an unfair coin which tosses head and tail with probability 0.01 and 0.99, respectively.

1. Toss this coin independently, and let X be the first time that head appears. Compute the mean and variance of X .
 2. Toss this coin for 500 times independently. Use a Poisson limit theorem to compute (approximately) the probability that exactly 5 many head appears.

See
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$$(6) \quad f(x) = c(x+3)$$

$$1 = \int_{-2}^1 f(x) dx = \int_{-2}^1 c(x+3) dx$$

$$= c \left. \frac{1}{2}(x+3)^2 \right|_{x=-2}^1 = c \frac{1}{2} 4^2 - c \cdot \frac{1}{2} 1^2 = \frac{15c}{2}$$

$$1 = \frac{15c}{2} \Rightarrow c = \frac{2}{15}$$

$$(2) \quad Y = |X+1| \quad S_X = (-2, 1) \Rightarrow S_Y = [0, 2)$$

$$F_Y(y) = P(Y \leq y)$$

$$= P(|X+1| \leq y)$$

$$= P(-y \leq X+1 \leq y)$$

$$= P(-y-1 \leq X \leq y-1)$$

$$= \begin{cases} \int_{-y-1}^{y-1} \frac{2}{15} (x+3) dx & \text{if } 0 \leq y \leq 1 \\ \int_{-2}^{y-1} \frac{2}{15} (x+3) dx & \text{if } 1 < y < 2 \end{cases}$$

Then compute $F_Y(y)$ in $0 \leq y \leq 1$ and $1 < y < 2$,

and differentiate wrt. y to get $f_Y(y)$.

